

# Global Valuation

## Cross-Country Valuation Multiples: Data and First Results

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February 2026

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# Outline

Data

Cross-Country Regressions

Leverage Across Countries

Variance Decomposition

Country Effects Over Time

Regional Trends

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Next Steps

# Data

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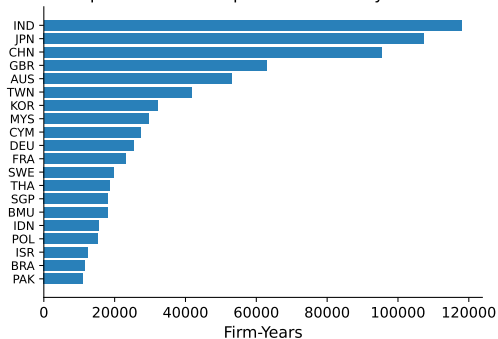
# Datasets

| File              | Size   | Rows  | Cols | Description                                |
|-------------------|--------|-------|------|--------------------------------------------|
| compustat_global  | 2.3 GB | 981K  | 468  | Global annual fundamentals (124 countries) |
| compustat_america | 0.8 GB | 254K  | 984  | NA fundamentals + CRSP link                |
| returns_global    | 1.9 GB | 10.4M | 25   | Global monthly prices                      |
| returns_america   | 0.9 GB | 3.6M  | 55   | CRSP US monthly returns                    |

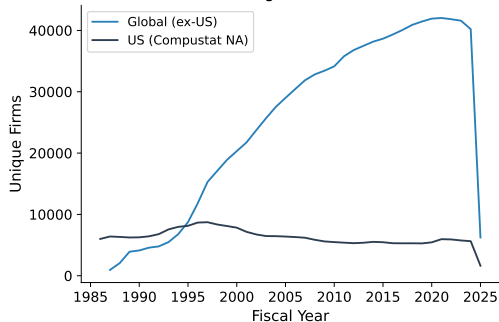
- Compustat Global: fiscal years 1987–2024, 124 countries
- Compustat America: fiscal years 1986–2024, CRSP-linked
- Combined regression panel: **743,957 firm-years**, 72,787 firms, 81 countries

# Coverage: Countries and Time

Compustat Global: Top 20 Countries by Firm-Years



Coverage Over Time



India, Japan, China dominate non-US coverage. Panel peaks at ~41K firms (2022).

## Panel Construction

- **P/E**:  $\log(\text{Market Cap}/\text{Net Income})$  for profitable firms ( $NI > 0$ )
- **M/B**:  $\log(\text{Market Cap}/\text{Book Equity})$
- **Leverage**:  $(\text{Total Assets} - \text{Book Equity})/\text{Total Assets}$
- **ROA**:  $\text{Net Income} / \text{Total Assets}$
- **Industry**: GICS Industry Group ( $\sim 25$  categories)
- Winsorize P/E, M/B, ROA, leverage at 1st/99th percentiles

| Variable           | N       | Mean  | P10   | Median | P90  |
|--------------------|---------|-------|-------|--------|------|
| $\log(\text{P/E})$ | 532,028 | 2.96  | 1.77  | 2.80   | 4.35 |
| $\log(\text{M/B})$ | 743,957 | 0.46  | -0.77 | 0.39   | 1.76 |
| Leverage           | 743,957 | 0.49  | 0.14  | 0.49   | 0.85 |
| ROA                | 743,523 | -0.01 | -0.15 | 0.02   | 0.11 |

# Cross-Country Regressions

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## Regression Specifications

**P/E** (profitable firms,  $NI > 0$ ):

$$\log(P/E)_{i,t} = \beta_1 \text{Lev}_{i,t} + \beta_2 \log(\text{Assets})_{i,t} + \beta_3 \Delta E_{i,t-1} + \gamma_c + \delta_j + \tau_t + \varepsilon_{i,t}$$

**M/B** (all firms):

$$\log(M/B)_{i,t} = \beta_1 \text{ROA}_{i,t} + \beta_2 \text{Lev}_{i,t} + \beta_3 \log(\text{Assets})_{i,t} + \beta_4 \Delta E_{i,t-1} + \gamma_c + \delta_j + \tau_t + \varepsilon_{i,t}$$

**ROA:**

$$\text{ROA}_{i,t} = \beta_1 \text{Lev}_{i,t} + \beta_2 \log(\text{Assets})_{i,t} + \gamma_c + \delta_j + \tau_t + \varepsilon_{i,t}$$

$\gamma_c$  = country FE,  $\delta_j$  = GICS industry FE,  $\tau_t$  = year FE. Standard errors clustered at the country level. Spec 2 replaces separate  $\gamma_c + \tau_t$  with  $\gamma_{c,t}$  (country×year FE).



## P/E Regression Results

|                  | <b>Spec 1</b>        | <b>Spec 2</b>        |
|------------------|----------------------|----------------------|
|                  | C + Ind + Year FE    | C×Year + Ind FE      |
| Leverage         | −0.017<br>(0.105)    | −0.014<br>(0.102)    |
| log(Assets)      | −0.068**<br>(0.030)  | −0.070**<br>(0.030)  |
| Lag Earn. Growth | −0.016***<br>(0.003) | −0.016***<br>(0.003) |
| N / $R^2$        | 428,722 / 0.133      | 428,722 / 0.154      |

Sample restricted to profitable firms ( $NI > 0$ ). Leverage insignificant; larger firms have lower P/E; past earnings growth *reduces* current P/E (mean reversion).

## M/B Regression Results

|                  | Spec 1               | Spec 2               |
|------------------|----------------------|----------------------|
|                  | C + Ind + Year FE    | C×Year + Ind FE      |
| ROA              | 0.232<br>(0.160)     | 0.198<br>(0.156)     |
| Leverage         | 0.911***<br>(0.064)  | 0.910***<br>(0.067)  |
| log(Assets)      | −0.068***<br>(0.026) | −0.071***<br>(0.027) |
| Lag Earn. Growth | 0.008***<br>(0.002)  | 0.007***<br>(0.002)  |
| N / $R^2$        | 593,530 / 0.211      | 593,530 / 0.237      |

Leverage has *huge* positive effect on M/B (+0.91). ROA insignificant after controls. M/B = Market Equity / Book Equity, not including debt.

## P/E vs M/B: Key Regression Differences

- **Leverage:** Insignificant for P/E ( $-0.02$ ) but *dominant* for M/B ( $+0.91^{***}$ )
  - M/B mechanically increases with leverage: less book equity  $\Rightarrow$  higher market/book
  - P/E is unaffected: price and earnings both scale with equity
- **Earnings growth:** Negative for P/E ( $-0.016$ ), positive for M/B ( $+0.008$ )
  - P/E: past earnings surge raises denominator  $\Rightarrow$  lower P/E
  - M/B: past growth signals quality  $\Rightarrow$  higher market premium over book
- **ROA:** Insignificant in both — after country and industry FE, current profitability does not predict cross-sectional valuation
- $R^2$ : P/E (0.133) < M/B (0.211) — M/B partly driven by leverage mechanics

## Leverage Across Countries

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# Leverage Varies Widely Across Countries

## Highest median leverage:

Tunisia 0.722

Portugal 0.722

Italy 0.660

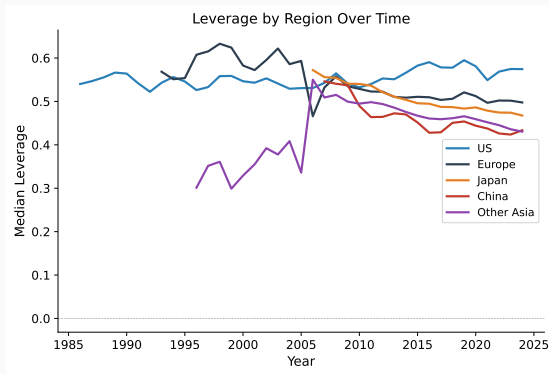
Spain 0.627

France 0.593

Australia 0.254

**Lowest:** Romania 0.242

Guernsey 0.218

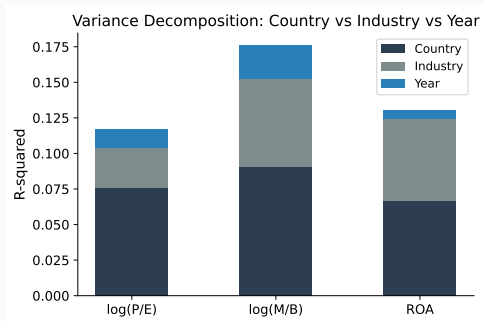


Cross-country std = 0.095, range = 0.50. Southern Europe highly levered; Anglo-Saxon countries lower. This heterogeneity makes M/B harder to compare across countries.

# Variance Decomposition

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# What Drives Valuation? Country vs Industry vs Year

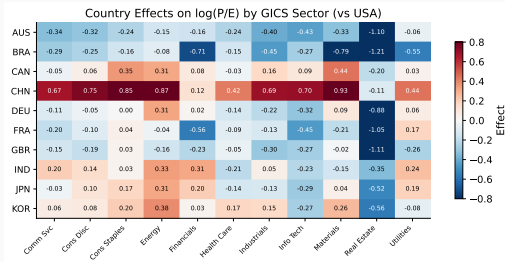


## Shapley $R^2$ decomposition:

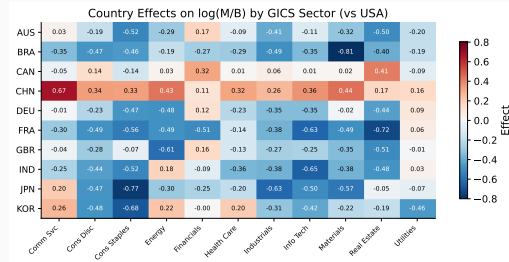
|          | P/E          | M/B   | ROA   |
|----------|--------------|-------|-------|
| Country  | <b>64.6%</b> | 51.6% | 50.9% |
| Industry | 24.3%        | 35.1% | 44.7% |
| Year     | 11.1%        | 13.3% | 4.4%  |

Country dominates all three. P/E shows *strongest* country dependence — industry matters more for M/B (partly via leverage norms).

# Industry × Country Interaction



P/E heatmap



M/B heatmap

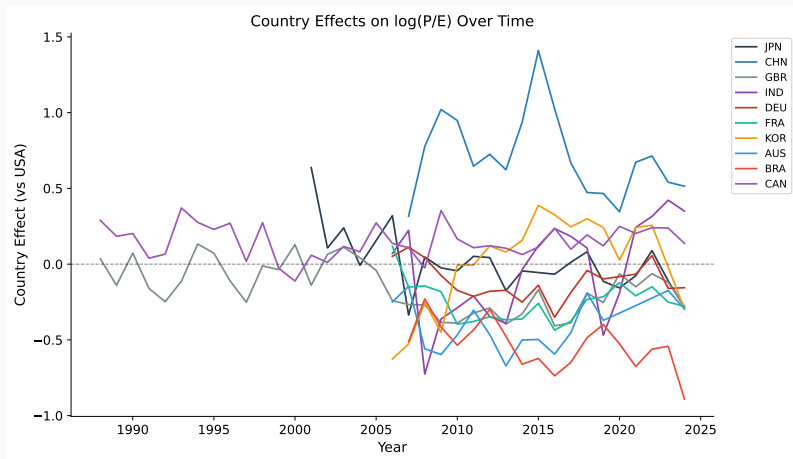
Country effects by GICS sector (vs USA). Real estate stands out with large negative effects in both metrics across most countries.



## Country Effects Over Time

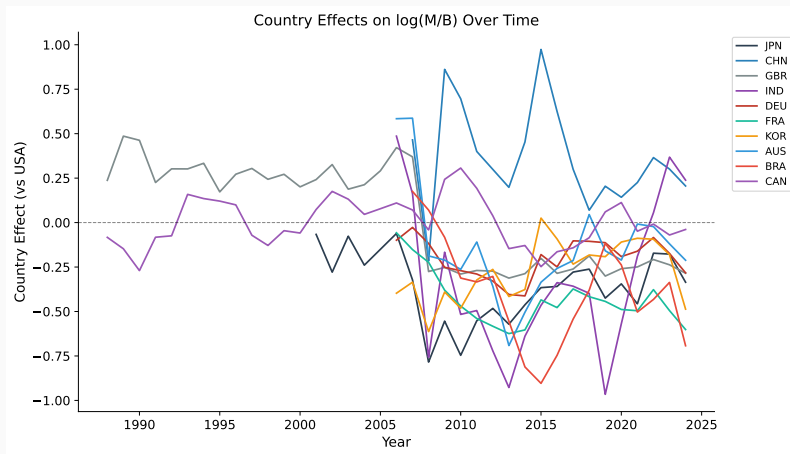
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# Country Effects on P/E (vs USA)



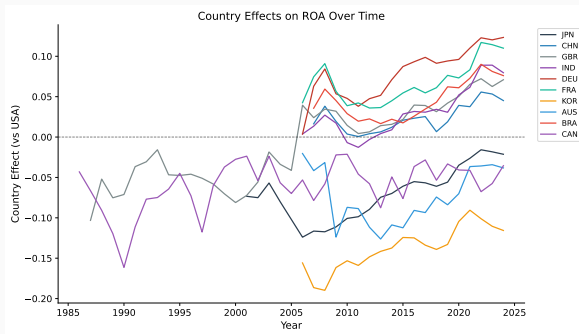
Year-by-year:  $\log(P/E) \sim \text{Leverage} + \log(\text{Assets}) + \Delta E_{t-1} + \text{Country FE} + \text{Industry FE}$ . USA = 0.

# Country Effects on M/B (vs USA)



Year-by-year:  $\log(M/B) \sim \text{Leverage} + \log(\text{Assets}) + \Delta E_{t-1} + \text{Country FE} + \text{Industry FE}$ . USA = 0.

# Japan: The Revealing P/E vs M/B Comparison



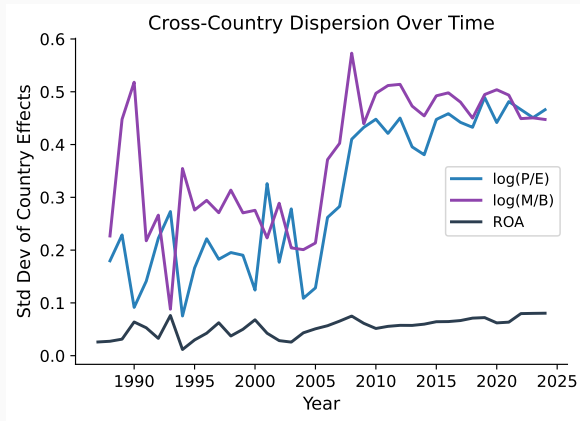
Japan's “discount” depends on the metric:

- M/B:  $-0.36$  (negative 100% of years)
- P/E:  $+0.01$  (near zero)

Japan's M/B discount reflects *conservative book values*, not lower market valuations vs earnings.

P/E is a cleaner cross-country comparison.

# Cross-Country Dispersion Over Time



## Dispersion is diverging:

- $P/E$ :  $0.17 \rightarrow 0.46$
- $M/B$ :  $0.40 \rightarrow 0.45$

$P/E$  dispersion increased *dramatically* ( $\times 2.7$ ) while  $M/B$  was more stable.

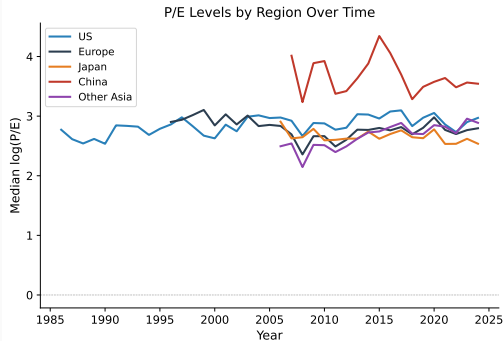
**ROA dispersion** more stable ( $\sim 0.05$ – $0.10$ ), with a spike during 2009–10 GFC.

Valuations are *diverging* despite financial globalization.

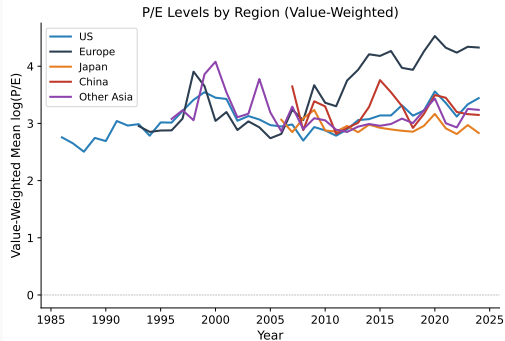
## Regional Trends

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# Regional Valuations: P/E



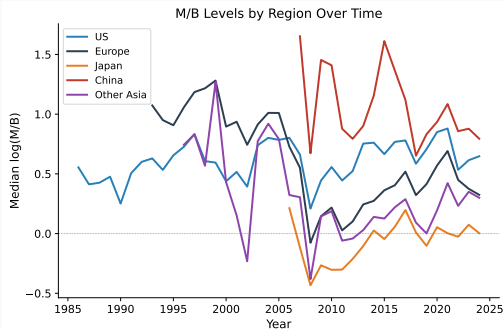
EW median



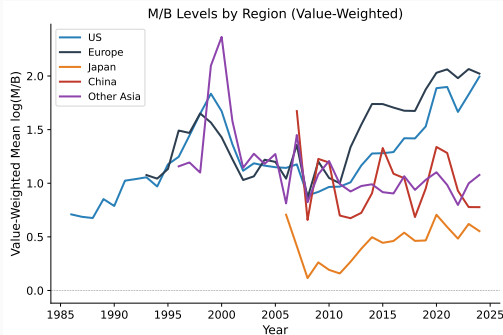
Value-weighted

China stands out with the highest P/E. Japan's P/E is comparable to Europe and Other Asia.

# Regional Valuations: M/B



EW median

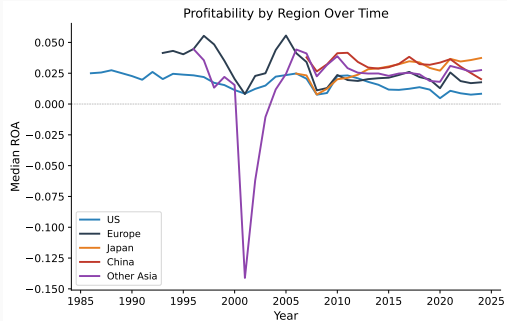


Value-weighted

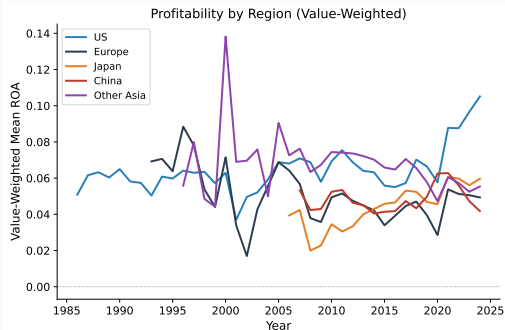
Japan's deep M/B discount disappears in P/E — book equity measurement drives the gap.



# Regional Profitability Over Time



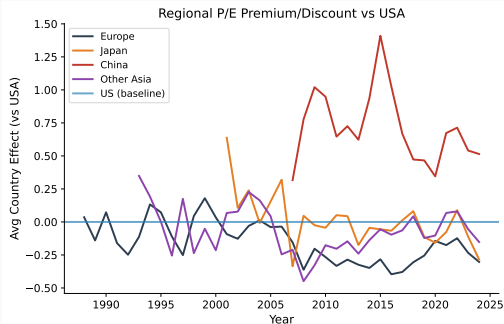
EW median



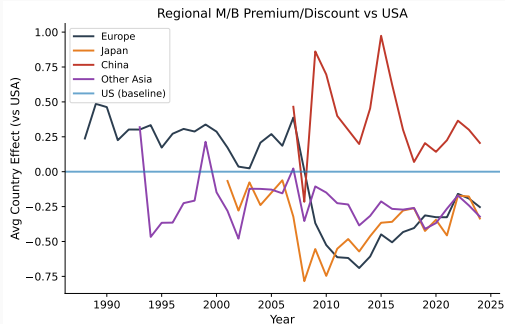
Value-weighted

Japan and Other Asia have *higher* profitability than US despite lower valuations.

# Regional Valuation Gap vs USA



P/E effects



M/B effects

Both metrics show Europe and Japan below US. China P/E premium is much larger than China M/B premium.

## The US Valuation Premium (Recent 5 Years)

| US minus   | P/E gap<br>(EW median) | M/B gap<br>(EW median) |
|------------|------------------------|------------------------|
| Europe     | +12%                   | +25%                   |
| Japan      | +36%                   | +98%                   |
| China      | −48%                   | −18%                   |
| Other Asia | +6%                    | +49%                   |

- Japan gap: +36% in P/E but +98% in M/B — **half the Japan gap is book equity measurement**
- China: high P/E (−48%) but modest M/B (−18%) — earnings are high relative to price
- Europe gap is moderate in both metrics

## Stylized Facts and Takeaways

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## Are P/E and M/B Results Aligned?

**Broadly yes** (country-level correlation = +0.85), **but with key differences:**

| Finding                 | P/E          | M/B          |
|-------------------------|--------------|--------------|
| Country share of $R^2$  | 64.6%        | 51.6%        |
| Industry share of $R^2$ | 24.3%        | 35.1%        |
| Leverage coefficient    | -0.02 (ns)   | +0.91***     |
| Japan effect (avg)      | +0.01        | -0.36        |
| Dispersion trend        | $\times 2.7$ | $\times 1.1$ |

**Key insight:** M/B is contaminated by leverage norms and book equity accounting. P/E is a *cleaner* cross-country comparison, revealing even stronger country dependence.

## Stylized Facts (1/2): Decomposition

1. **Country effects dominate valuation** — country explains 52–65% of  $R^2$  (Shapley), more in P/E than M/B. Extends Heston–Rouwenhorst from *returns* to *levels*.
2. **Country dominates profitability too** — for ROA, country edges industry (51% vs 45%).
3. **Negative valuation–ROA correlation** ( $\rho \approx -0.37$ ) — high-valuation countries are *not* high-ROA countries. Consistent across both P/E and M/B.
4. **M/B is leverage-contaminated** — leverage coefficient  $+0.91^{***}$  in M/B, insignificant in P/E. Cross-country leverage heterogeneity (std = 0.095) mechanically inflates M/B variation.

## Stylized Facts (2/2): Dynamics

5. **Japan discount is a measurement artifact** — M/B effect  $-0.36$  (100% negative years), but P/E effect  $\approx 0$ . Conservative Japanese book equity accounting drives the M/B gap.
6. **China premium is real but declining** — high in both P/E and M/B, but falling since 2015.
7. **Valuation dispersion is diverging** — P/E dispersion nearly tripled ( $0.17 \rightarrow 0.46$ ). Despite globalization, country valuations are *not converging*.
8. **Profitability does not explain valuations** — ROA coefficient is insignificant in both P/E and M/B regressions after country/industry FE. Discount rates and growth expectations are the key channels.

# Literature

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# Related Literature

## Country vs industry effects:

- Heston–Rouwenhorst (1994, *JFE*) — country dominates returns. We extend to *levels*.
- Griffin–Karolyi (1998) — industry effects vary across sectors.

## Institutions and investor protection:

- La Porta et al. (2002, *JF*) — investor protection raises Tobin's Q.
- Hail–Leuz (2006, *JAR*) — institutions reduce cost of equity by up to 200 bps.
- Stulz (2005, *JF*) — “twin agency problems” prevent equalization.

## Market-to-book as a metric:

- Penman et al. (2007, *JAR*) — decompose M/B into leverage and operating components.
- Our P/E vs M/B divergence on Japan echoes this concern.

## Next Steps

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## Research Questions Raised

1. **Why is P/E dispersion diverging?** ( $\times 2.7$ ) — US tech sector? Post-GFC monetary policy? Institutional changes?
2. **What explains China's P/E premium?** — Highest P/E despite moderate M/B. Retail investors? Capital controls? Growth expectations?
3. **Can we decompose the discount rate channel?** — Country FE mix risk premia, liquidity, governance, investor access
4. **Why are valuations negatively correlated with ROA?** ( $\rho \approx -0.37$ ) — Capital allocation inefficiency? Higher risk in profitable countries?
5. **Japan as natural experiment** —  $P/E \approx 0$  vs  $M/B = -0.36$ : can we bound the “accounting noise” in M/B?

## Suggested Next Steps

1. **EV/EBITDA** as third metric — avoids book equity issues (M/B) and profitable-firms restriction (P/E)
2. **Institutional variables** — legal origin, anti-self-dealing index, capital controls, sovereign ratings as country-level regressors
3. **Realized returns** — do low-valuation countries earn higher returns? Link to Bekaert–Harvey, Fama–French international
4. **Persistence** — Markov transition matrices for country valuation quintiles: permanent or mean-reverting?
5. **Sector decomposition** — Real Estate shows extreme country effects. Tradable vs non-tradable sectors
6. **Within-country dispersion** — does country-level valuation heterogeneity predict aggregate returns?